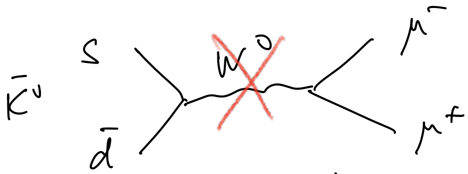
No lecture tuesday 20 May

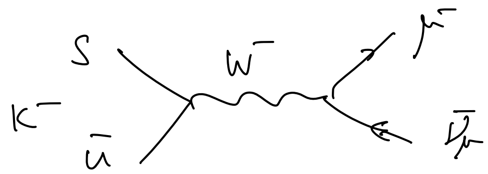
Charged weak current — Flavor Changing

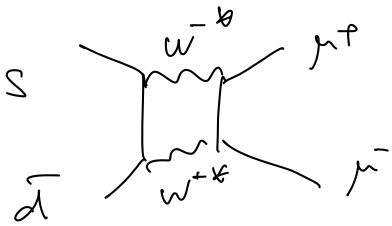
$V_{ij} \in V_{CKM}$

1) $K^0 \to \mu^+\mu^- \simeq 10^{-9} \neq \emptyset$

2) $K^+ \to \mu^+\nu_\mu \gg K^0 \to \mu^+\mu^-$

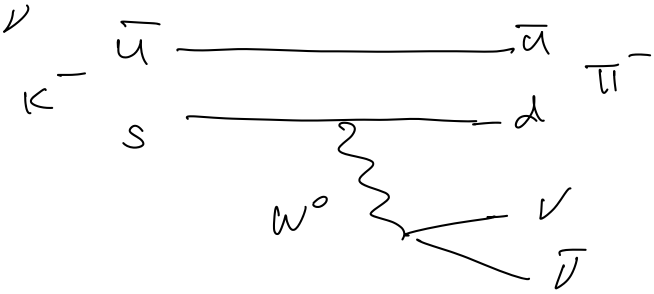






3) $K^+ \to \pi^+ \nu\bar{\nu}$

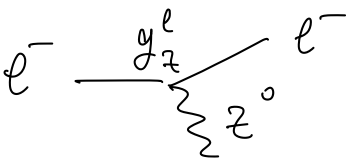
suppressed

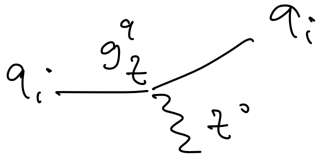


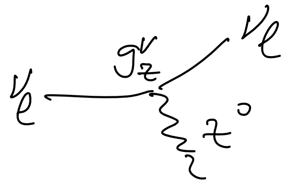
exp evidence $\Rightarrow$ No flavor changing neutral current (W^0)

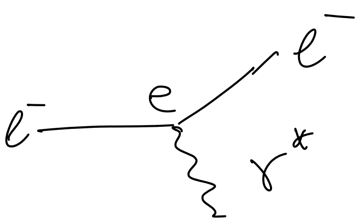
$\Rightarrow$ what about Flavor-conserving neutral current?

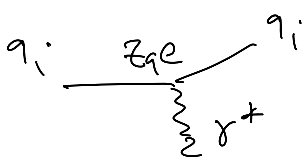
Neutral Weak Current

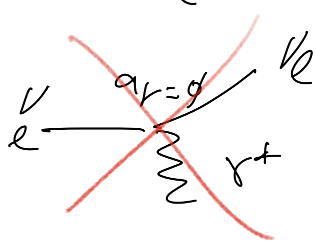






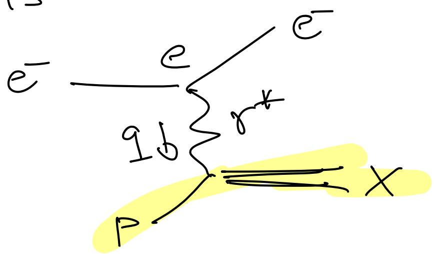


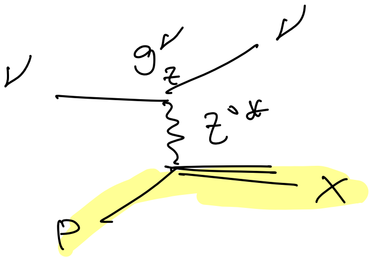




NO QED equivalent

DIS



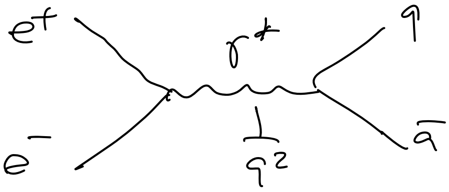


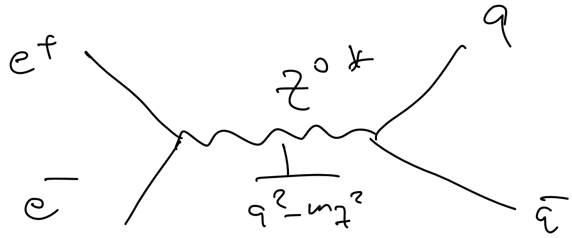
t-channel diagram.

$$M_\gamma \sim \frac{e}{q^2} \qquad\qquad M_Z \sim \frac{g_Z^\nu}{q^2 - m_Z^2}$$

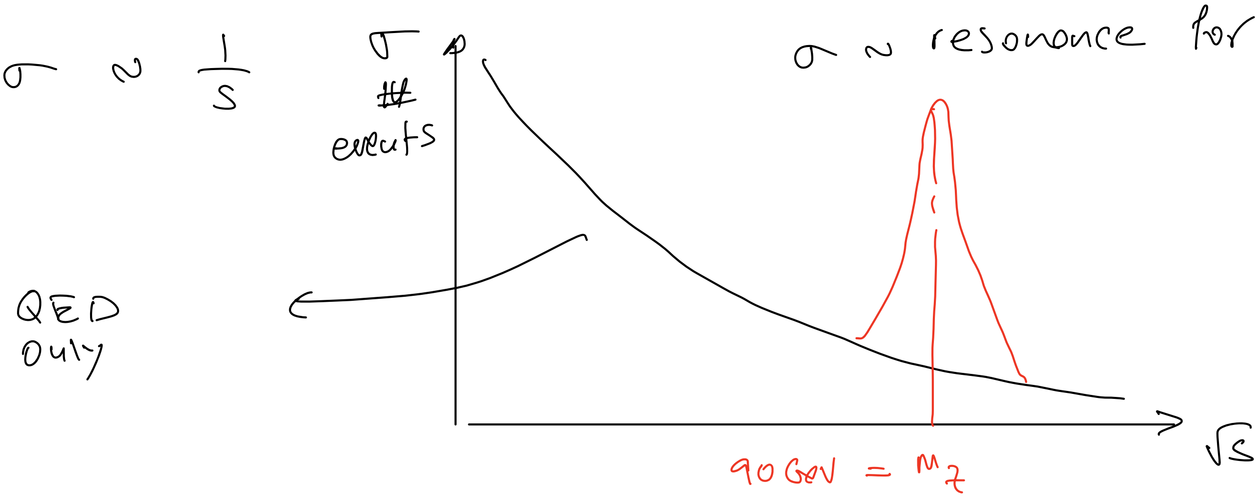
For $q^2 \ll m_Z^2$ $\quad |M_\gamma|^2 \gg |M_Z|^2 \simeq \dfrac{g_Z^2}{m_Z^4}$

you notice Z^0 for $q^2 \simeq m_Z^2$



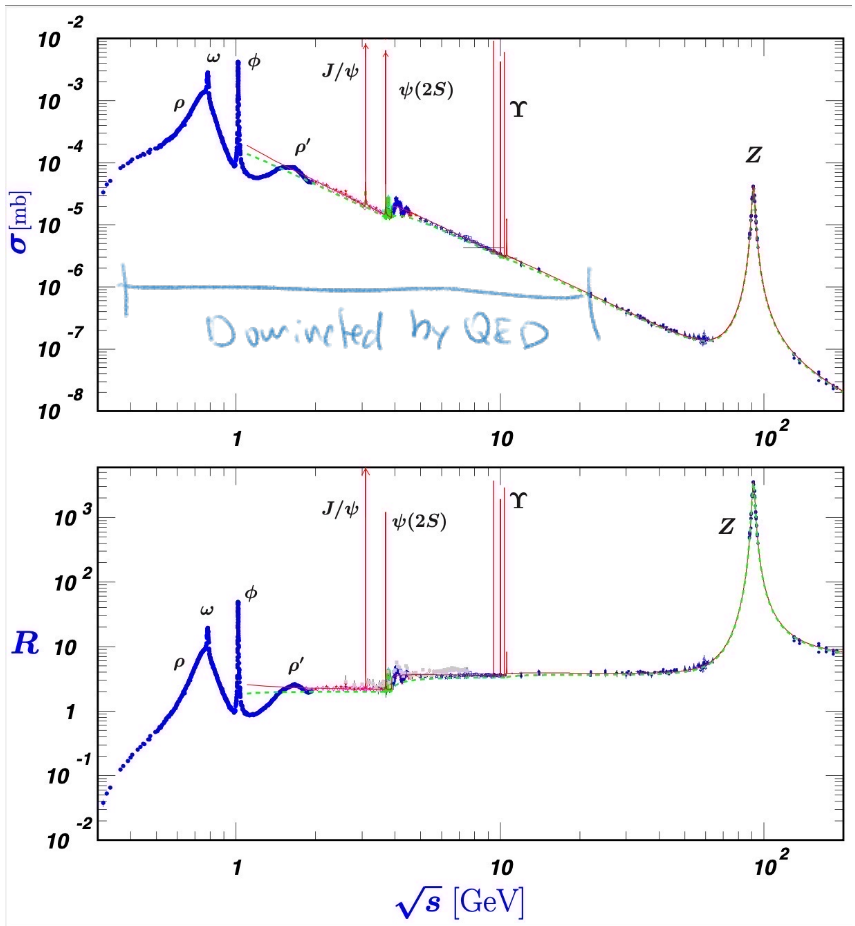


$\sigma \sim \frac{1}{s}$ $\qquad\qquad$ $\sigma \sim$ resonance for $q^2 \simeq m_Z^2$



$\sqrt{s} = 90$ GeV $\Rightarrow$ $E_{beam} = 45$ GeV

LEP $\quad$ 1989 – 2002 $\quad$ $\sqrt{s} \simeq 90$ GeV $\to$ 200 GeV



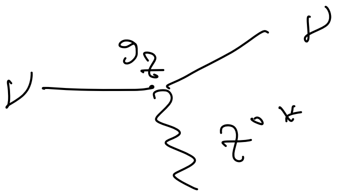
Direct proof of Z

Requires knowledge of m_Z

$\Rightarrow$ set properly $E_{beam} = \frac{m_Z}{2}$

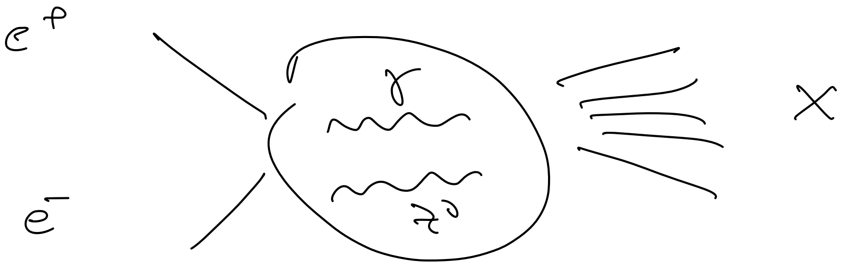
otherwise use hadron colliders.

$\Rightarrow$ t-channel provides indirect proof of Z^0



no background from QED.

1961: Glashow proposed unification of EM and weak int. if Z^0 exists: exp. see a mixture of Z^0 and γ

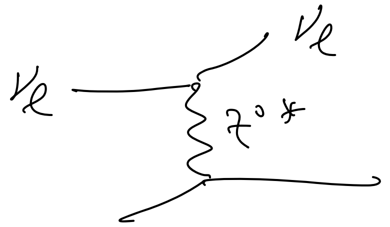


$$\sigma \propto |M|^2 = |M_\gamma + M_Z|^2 = |M_\gamma|^2 + |M_Z|^2 + \underbrace{M_\gamma^\dagger M_Z + M_Z^\dagger M_\gamma}_{\text{interference}}$$

1967: Weinberg and Salam: unification of EM + weak as a spontaneous symm. breaking of $SU(2)_{weak} \times U(1)_{EM}$.

1971: t'Hooft proved that unified theory is renormalizable.

Indirect evidence for Z^0



$q_\nu = \emptyset \Rightarrow$ NO QED

No color $\Rightarrow$ NO QCD

$\Rightarrow$ if events observed.

$\Rightarrow$ flavor conserving neutral weak current

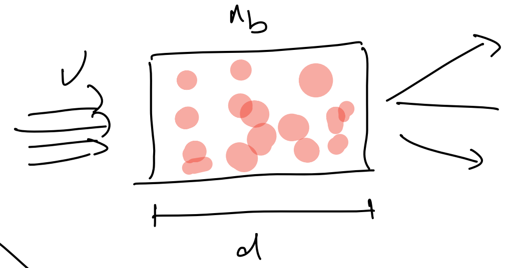
Challenges:

1) beam of neutrinos.

2) $\bar{\nu}_e + p \to n + e^+$ (Reiner-Cowan exp. 1956)

$$\sigma \sim 10^{-43}\ \text{cm}^2 \times \left[\frac{p}{\text{MeV}}\right]^2$$

3) high intensity + focused beam



$$\frac{dN_r}{dt} = \sigma_\nu \cdot \frac{dN_r}{dt} \cdot \underbrace{n_b}_{\text{very thick material.}} \cdot \underbrace{d}_{\text{as big as possible.}}$$

Exp. @ CERN 1973

$\pi^\pm \to \mu^\pm\, \nu_\mu / \bar{\nu}_\mu \quad \simeq 100\,\%$

$\ e^\pm\, \nu_e / \bar{\nu}_e \quad \simeq 10^{-4}\,\%$

$K^\pm \to \mu^\pm\, \nu_\mu / \bar{\nu}_\mu \quad 64\,\%$

$K^\pm \to \pi^0\, \mu^+\, \nu_\mu$

$\ \pi^0\, e^+\, \nu_e$

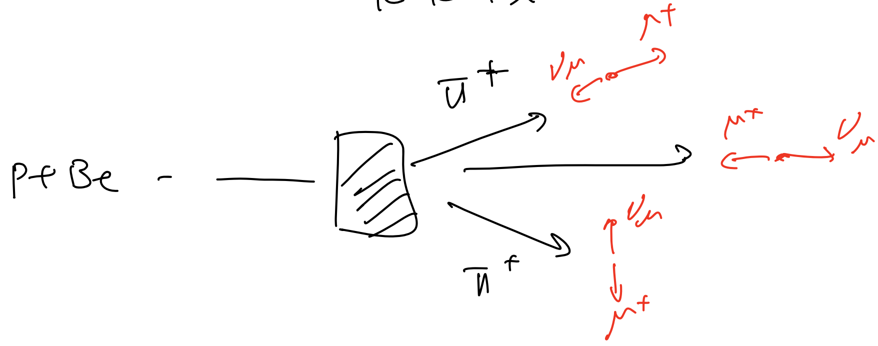
K^+ DECAY MODES	Fraction (Γ_i/Γ)	Scale factor/ Confidence level	p (MeV/c)
Leptonic and semileptonic modes			
$e^+ \nu_e$	(1.582 ± 0.007) $\times 10^{-5}$		247
$\mu^+ \nu_\mu$	(63.56 ± 0.11) %	S=1.2	236
$\pi^0 e^+ \nu_e$	(5.07 ± 0.04) %	S=2.1	228
Called K^+_{e3}.			
$\pi^0 \mu^+ \nu_\mu$	(3.352 ± 0.033) %	S=1.9	215
Called $K^+_{\mu 3}$.			
$\pi^0 \pi^0 e^+ \nu_e$	(2.55 ± 0.04) $\times 10^{-5}$	S=1.1	206
$\pi^+ \pi^- e^+ \nu_e$	(4.247 ± 0.024) $\times 10^{-5}$		203
$\pi^+ \pi^- \mu^+ \nu_\mu$	(1.4 ± 0.9) $\times 10^{-5}$		151
$\pi^0 \pi^0 \pi^0 e^+ \nu_e$	$<$ 3.5 $\times 10^{-6}$	CL=90%	135
Hadronic modes			
$\pi^+ \pi^0$	(20.67 ± 0.08) %	S=1.2	205
$\pi^+ \pi^0 \pi^0$	(1.760 ± 0.023) %	S=1.1	133
$\pi^+ \pi^+ \pi^-$	(5.583 ± 0.024) %		125

obtain pure beam of $\nu_\mu / \bar{\nu}_\mu$ and reject $\nu_e / \bar{\nu}_e$

$p + Be \rightarrow \pi^+ + X$
$\pi^- + X$
$K^+K^- + X$

In π^+ reference

$\mu^+ \longleftarrow \bullet \longrightarrow \nu_\mu$



Beam of $\overset{(-)}{\nu}_\mu$ not focused and momenta changing a lot

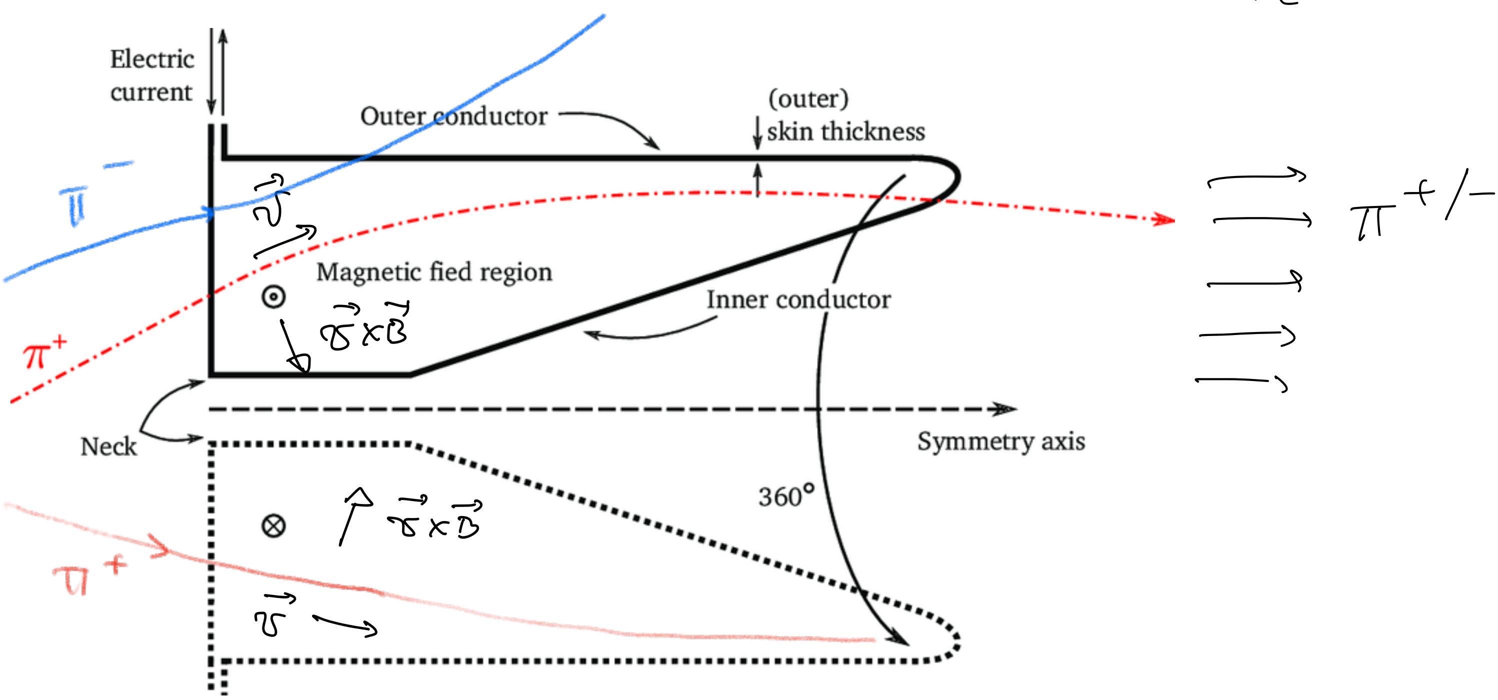
B field selects π^+/π^- and K^+/K^-

$\Rightarrow$ choose between ν_μ, $\bar{\nu}_\mu$

Exp. problems: focus $\pi^\pm$ in final state.

Magnetic Horn by Simon van der Meer
based on Lorentz Force

$\vec{F}_L = q\, \vec{v} \times \vec{B}$



Focus $\pi^{+/-}$ $\Longrightarrow$ Focus $\nu_\mu / \bar{\nu}_\mu$ beam.

$$\mu^+ \longleftarrow \overset{\pi^+}{\circ} \longrightarrow \nu_\mu$$

$$Q = m_\pi - m_\mu = 135 - 106 \text{ MeV} \simeq 29 \text{ MeV}$$

$$E_\pi = m_\pi = E_\nu + E_\mu = p^* + \sqrt{p^{*2} + m_\mu^2}$$

$$p^* = \frac{1}{2m_\pi}(m_\pi^2 - m_\mu^2) \simeq 25 \text{ MeV}$$

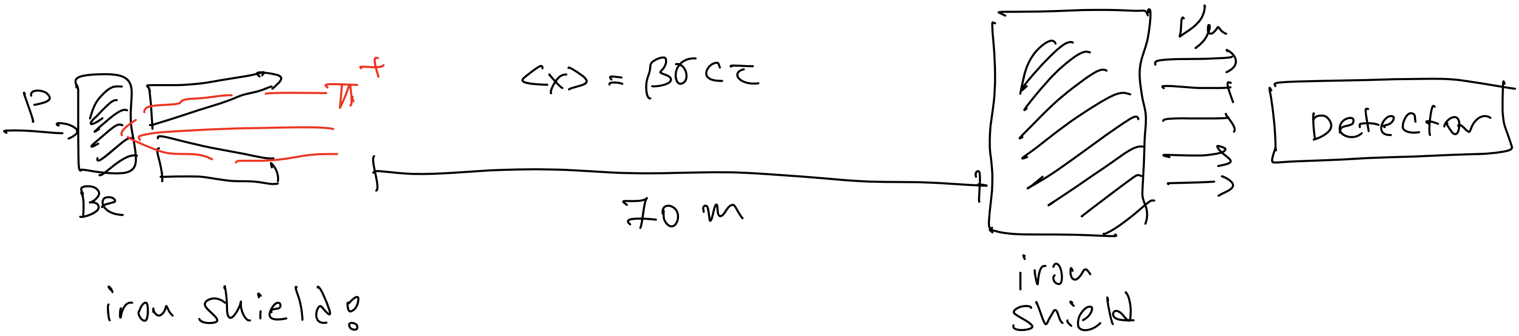
$$\beta\gamma = \frac{p^*}{m_\mu} = \frac{25}{106} \simeq 0.25$$

Reactor β^- decays: $E_\nu \simeq 2$ MeV

In collisions: $E_\nu^* \simeq 25$ MeV $\qquad E_\nu^{LAB} = \gamma_\pi (E_\nu^* + \beta_\pi p_\parallel^*)$

$$\sigma_{\nu+X} \simeq 10^{-43} \text{ cm}^2 \times \left(\frac{p}{\text{MeV}}\right)^2$$

$$\langle x \rangle = \beta\gamma c\tau$$

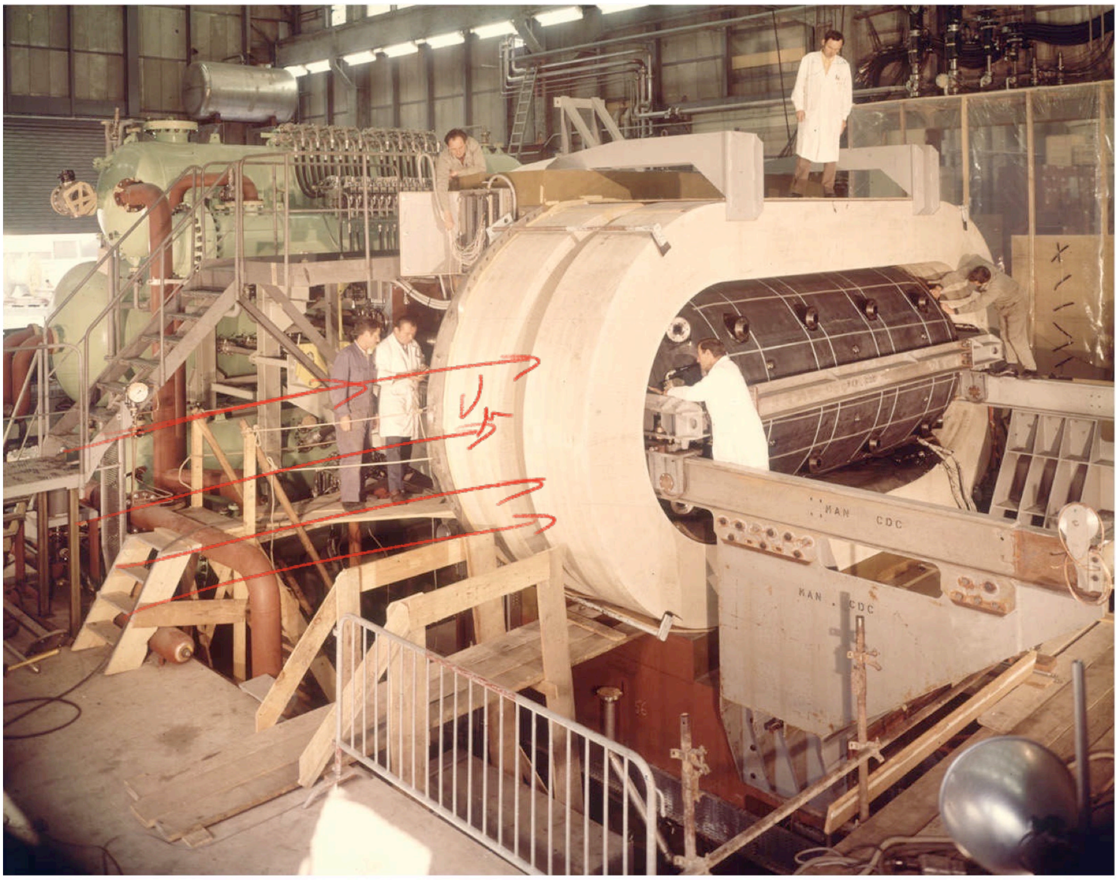


iron shield:

π, K, hadrons: nuclear interaction, ionization

$e^\pm$: EM shower, ionisation

$\mu^\pm$: ionization not stopped completely.



Gergemelle

Steel cylinder filled w/ Freon CF_3Br (liquid)

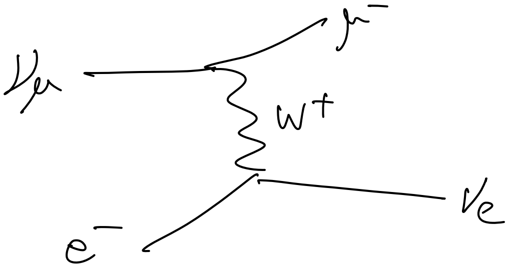
Bubble Chamber

$B = 2T$

$$\nu_\mu + \left\{ \begin{matrix} e^- \\ p \\ n \end{matrix} \right) \longrightarrow \nu_\mu + X$$ neutral current

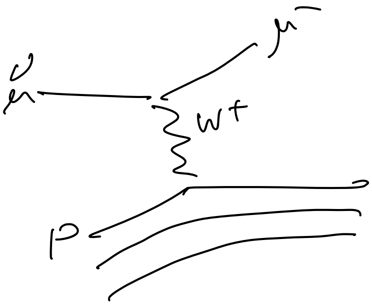
main background: Charged weak current w/ $W^\pm$

Charged Weak Current (CC)



$\nu_\mu + e^- \rightarrow \nu_e + \mu^-$

$\nu_\mu + n \rightarrow \mu^- + p/X$



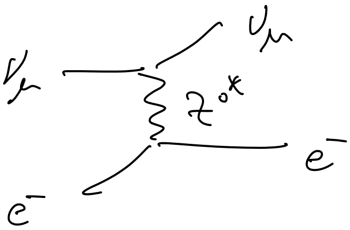
$\nu_\mu + p \rightarrow \Delta^{++} + X \quad + \mu^-$

$\pi^+ + X$

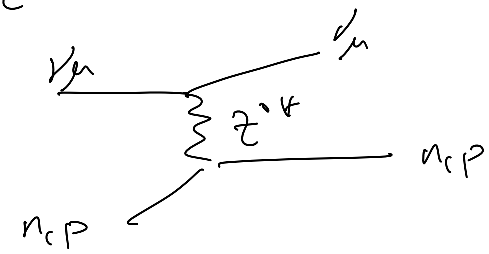
hadrons

presence of μ^- in final state $\Rightarrow$ CC interaction (background).

If NC exists $\Rightarrow$ additional processes.

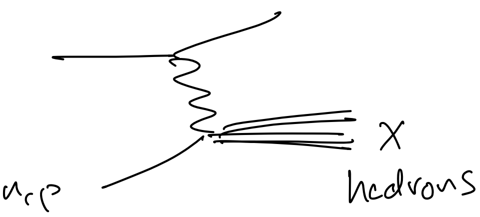


$\nu_\mu + e^- \rightarrow \nu_\mu + e^-$



$\nu_\mu + n/p \rightarrow \nu_\mu + n/p$

elastic scattering.



$\nu_\mu + n/p \rightarrow \nu_\mu + $ hadrons

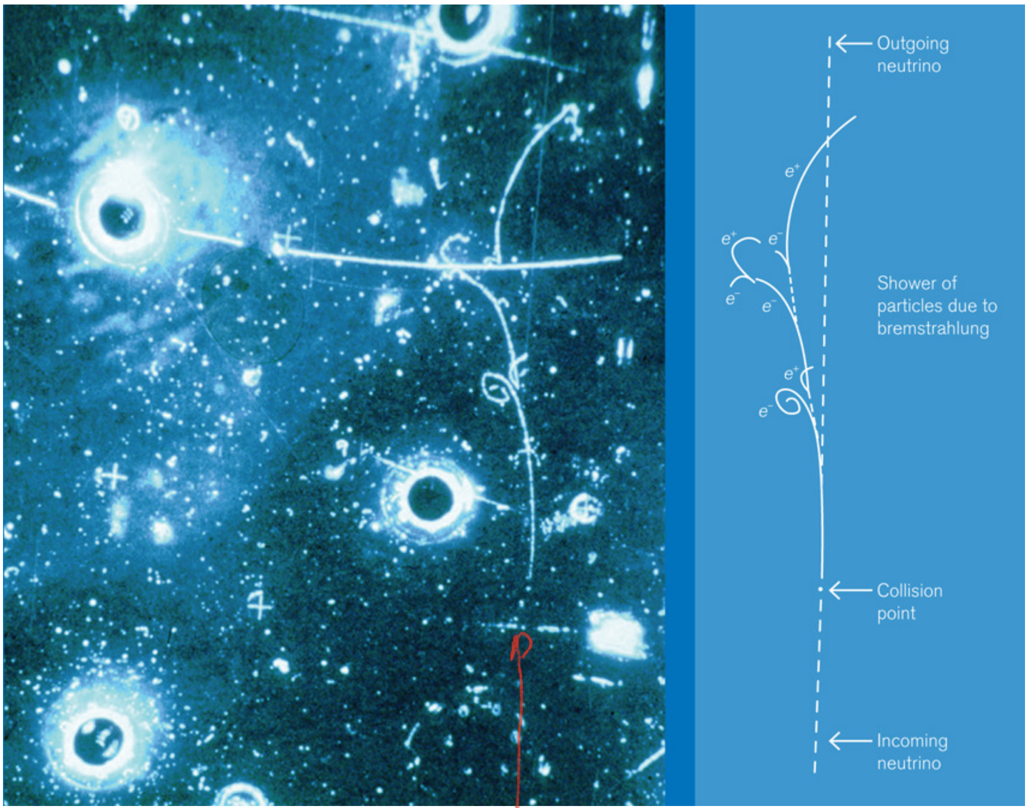
inelastic

ν_μ $\vec{P}_{\nu_\mu}$ CC μ^- Charged particles. $\sum_i \vec{P}_{T_i} = 0$ Balanced

$\vec{P}_{\nu_\mu}$ NC $\vec{P}_{\nu_\mu'}$ $\sum_i^{\text{charged}} \vec{P}_{T_i} \neq 0$

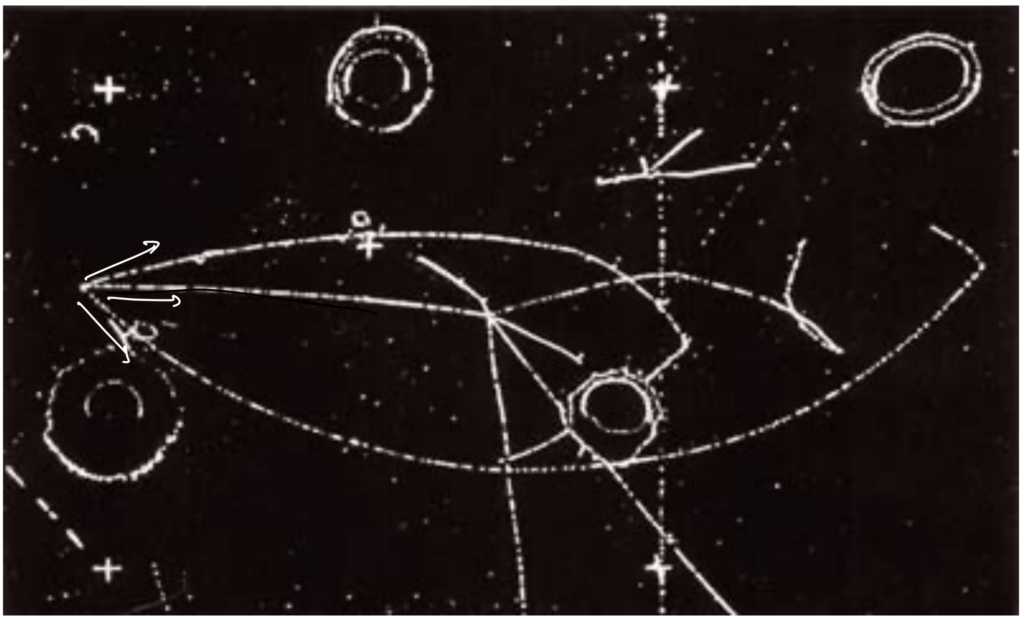
$\vec{P}_{T_{\nu'}} = - \sum_i^{\text{Charged}} \vec{P}_{T_i}$

unbalanced.



$\nu_\mu + e^- \to \nu_\mu + e^-$

ν_μ



$\nu_\mu + n/p \to$ hadrons $+ \nu_\mu$

unbalanced hadronic event

no muons

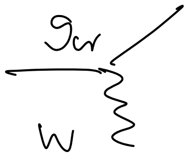
$\pi^+ \to \nu_\mu$ beam : 102 NC, 428 CC, 15 n (neutron background)

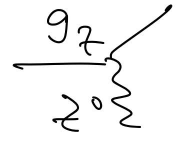
$\pi^- \to \bar{\nu}_\mu$ beam: 64 NC, 148 CC, 12 n (neutron background)

Conclusions

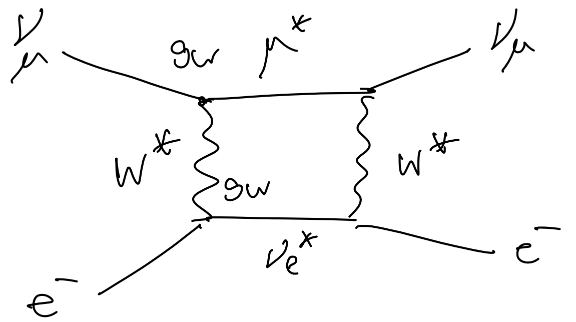
1) $\frac{\# NC}{\# CC} \neq 0 \Rightarrow$ NC exists

2) $\frac{\# NC}{\# CC} \sim \frac{1}{3} \Rightarrow g_W \neq g_Z$





3) $\frac{1}{3} >>$ no Z^0 hypothesis
not compatible with higher order W-only process



2^{nd} order CC-only process.

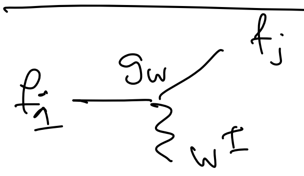
$$M_W \simeq \left(\frac{g_W^2}{M_W^2}\right)^2 \sim G_F^2$$

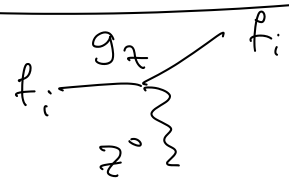
$$\frac{NC}{CC} \simeq \frac{1}{3} \neq (G_F)^4$$

not compatible with observed $\# NC / \# CC$

$\Rightarrow$ indirect evidence for Z^0

Glashow - Weinberg - Salam theory of Electroweak (GWS)





$f_i = e^\pm, \nu_e, \bar{\nu}_e, q_j, \bar{q}_j$

j: flavor of quarks.

$i g_W \gamma^\mu (1-\gamma^5)$

$i g_Z \gamma^\mu (c_V - c_A \gamma^5)$

GWS: predicts values of c_V, c_A for all fermions

f	c_V	c_A
ν_e, ν_μ, ν_τ	$\frac{1}{2}$	$\frac{1}{2}$
e^-, μ^-, τ^-	$-\frac{1}{2} + 2\sin^2\theta_w$	$-\frac{1}{2}$
u, c, t	$\frac{1}{2} - \frac{4}{3}\sin^2\theta_w$	$\frac{1}{2}$
d, s, b	$-\frac{1}{2} + \frac{2}{3}\sin^2\theta_w$	$-\frac{1}{2}$

θ_W: weak mixing angle

Theory: $SU(2)_{weak} \times U(1)_{EM}$

3 bosons: W_1, W_2, W_3 $\qquad$ B $\qquad$ gauge fields

$$W_1, W_2 \longrightarrow (W_1 \pm i W_2) = W^{\pm}$$ physical charged vector bosons

$$\begin{pmatrix} Z^0 \\ A \end{pmatrix} = \begin{pmatrix} \cos\theta_w & -\sin\theta_w \\ \sin\theta_w & \cos\theta_w \end{pmatrix} \begin{pmatrix} W_3 \\ B \end{pmatrix}$$

θ_w: mixing angle between 2 neutral gauge fields

↳ physical particles.

Z^0: mass to be measured.

$A \equiv \gamma$: classical photon from EM

GWS: predicts g_w, g_e, g_Z

$$g_e = \frac{e}{\sqrt{4\pi}} = \sqrt{\alpha_{EM}} \qquad g_w = \frac{g_e}{\sin\theta_w} \qquad g_Z = \frac{g_e}{\sin\theta_w \cos\theta_w}$$

electric charge

Measure θ_w from weak processes.

$$\left(\frac{g_w^2}{m_w^2}\right)^2 \qquad \left(\frac{g_Z^2}{m_Z^2}\right)^2$$

$$\propto \qquad \propto$$

$$\#CC \qquad \#NC$$